Gout/CPPD — Diagnosis, MSU Crystals, Synovial Fluid, Imaging, Tophi

1. Blue needle crystal

WHAT YOU SEE

Glowing thought-bubble crystals, one labeled BLUE NEEDLE

WHAT IT ENCODES

Monosodium urate (MSU) crystals are NEEDLE-shaped and NEGATIVELY birefringent: they appear YELLOW when aligned parallel to the red compensator axis and BLUE when perpendicular. Mnemonic — 'the urate needle is yellow when parallel.' Identifying intracellular needles inside neutrophils on polarized microscopy confirms an acute gout flare.

BOARD TRAP

WRONG answer: calling MSU rhomboid and positively birefringent (blue when parallel). That describes CPPD (pseudogout) — the discriminator is needle vs rhomboid shape and the parallel-axis color (yellow = MSU gout, blue = CPPD).

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2. Microscope (gold standard)

WHAT YOU SEE

Detective at a microscope

WHAT IT ENCODES

The definitive diagnosis of gout is arthrocentesis with polarized light microscopy showing intracellular, negatively birefringent MSU needles in synovial fluid — this is the GOLD STANDARD and also samples for Gram stain/culture to exclude infection. Clinical criteria (e.g., ACR/EULAR) can support diagnosis when aspiration is not feasible, but crystal identification is definitive.

BOARD TRAP

WRONG answer: ruling out gout because the serum uric acid is normal. Serum urate can be NORMAL or even low during an acute flare (urate shifts into the joint/acute-phase effect), so a normal level does NOT exclude gout — the discriminator is joint-fluid crystal analysis, not the serum level.

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3. Polarized light

WHAT YOU SEE

Microscope slide labeled POLARIZED

WHAT IT ENCODES

Compensated polarized light microscopy is required to assess birefringence and tell MSU from CPPD. A red (first-order) compensator is inserted so the sign of birefringence can be read by crystal color relative to the slow axis: negative (MSU) = yellow when parallel, positive (CPPD) = blue when parallel. Without the compensator, you cannot reliably assign the sign.

BOARD TRAP

WRONG answer: trying to distinguish the crystals on ordinary (non-polarized) light or assuming any crystals seen are urate. You need compensated polarized microscopy to determine birefringence sign — the discriminator is the parallel-axis color under the red compensator, not crystal presence alone.

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4. Synovial fluid / inflammatory

WHAT YOU SEE

Slide and flare markers on the desk

WHAT IT ENCODES

Gout produces INFLAMMATORY synovial fluid: WBC roughly 2,000-50,000/microL, predominantly neutrophils, with reduced viscosity. This places it between non-inflammatory fluid (<2,000, e.g., osteoarthritis/apatite) and frankly septic fluid (often >50,000, frequently >100,000). Cell count guides but does not replace crystal analysis and culture.

BOARD TRAP

WRONG answer: assuming a high inflammatory cell count proves gout and excludes infection. Septic arthritis classically gives even higher counts and can COEXIST with crystals — the discriminator is positive Gram stain/culture, so a markedly elevated count still mandates ruling out sepsis.

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5. Aspirate!

WHAT YOU SEE

Wanted board with 'ASPIRATE!' and differentials SEPTIC/CPPD/PSORIASIS/CELLULITIS

WHAT IT ENCODES

Any acute monoarthritis should be ASPIRATED to exclude SEPTIC arthritis before being treated as gout, because untreated joint infection rapidly destroys cartilage. The differential for a hot, swollen joint also includes CPPD, psoriatic/reactive arthritis, and overlying cellulitis. Fluid is sent for cell count/differential, crystals, Gram stain, and culture.

BOARD TRAP

WRONG answer: starting empiric gout therapy in a hot joint without aspirating. Gout and septic arthritis can coexist, and missing infection is dangerous — the discriminator is the aspirate (Gram stain/culture), so never assume crystal disease without excluding sepsis.

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6. Gun = bacterium (culture)

WHAT YOU SEE

'TWO CRYSTALS' poster with a gun and 'CULTURE!'

WHAT IT ENCODES

Always send synovial fluid for Gram stain and CULTURE even when crystals are identified, because septic arthritis can occur simultaneously with crystal disease. A positive culture (often S. aureus, or N. gonorrhoeae in young sexually active patients) changes management to urgent drainage and antibiotics regardless of crystals present.

BOARD TRAP

WRONG answer: skipping culture once urate crystals are confirmed. Finding crystals does NOT rule out infection — coexisting septic arthritis is a classic trap, so the discriminator is always obtaining Gram stain and culture in addition to crystal analysis.

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7. Tophi (ear helix)

WHAT YOU SEE

Photo of ear helix labeled EAR HELIX

WHAT IT ENCODES

Tophi are chronic, firm subcutaneous deposits of MSU crystals seen in long-standing hyperuricemia/tophaceous gout. Classic locations include the HELIX of the ear, olecranon bursa, finger pads, toes, and the Achilles tendon. Their presence signals chronic poorly controlled disease and is an indication for urate-lowering therapy (e.g., allopurinol titrated to target serum urate <6 mg/dL, or <5 with tophi).

BOARD TRAP

WRONG answer: mistaking an ear-helix tophus for a rheumatoid nodule. Rheumatoid nodules occur over extensor surfaces (e.g., olecranon) with RA serology, whereas ear-helix deposits are highly specific for gouty tophi — the discriminator is location plus aspiration showing MSU crystals.

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8. Olecranon bursa tophus

WHAT YOU SEE

Elbow photo labeled OLECRANON

WHAT IT ENCODES

The olecranon and prepatellar bursae are common sites of tophaceous deposits and gouty bursitis in chronic disease. Aspiration of an olecranon bursa swelling can reveal chalky MSU material and confirm gout, while also distinguishing it from septic bursitis or a rheumatoid nodule.

BOARD TRAP

WRONG answer: assuming any olecranon swelling is a rheumatoid nodule or simple traumatic bursitis. In chronic gout it may be a tophus packed with MSU crystals — the discriminator is aspiration showing negatively birefringent needles and chalky-white material.

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9. Achilles tendon tophus

WHAT YOU SEE

Heel photo labeled ACHILLES TENDON

WHAT IT ENCODES

Urate can deposit in and around the Achilles tendon, producing tophi and a chronic tophaceous gout finding; tendon and entheseal deposition can cause pain and, rarely, rupture. It is part of the classic tophi distribution (ear helix, olecranon, fingers/toes, Achilles).

BOARD TRAP

WRONG answer: attributing posterior heel pain/swelling to isolated tendinopathy in a patient with known gout. Chronic gout can deposit urate in the Achilles tendon — the discriminator is the tophaceous context and crystal-proven MSU deposition rather than purely mechanical tendinitis.

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10. Ultrasound double-contour

WHAT YOU SEE

US image labeled DOUBLE WHITE LINE

WHAT IT ENCODES

The ultrasound 'double-contour sign' is a hyperechoic line of urate coating the surface of the hyaline cartilage, lying parallel to the bright subchondral bone — fairly SPECIFIC for gout and useful for early or intercritical diagnosis. Ultrasound can also show tophi and erosions noninvasively.

BOARD TRAP

WRONG answer: equating the cartilage-SURFACE double-contour of gout with the cartilage-WITHIN calcification (chondrocalcinosis) of CPPD. The discriminator is location: urate coats the cartilage surface (double contour = gout) whereas CPP calcifies within the cartilage substance (chondrocalcinosis = CPPD).

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11. Dual-energy CT (DECT)

WHAT YOU SEE

Colored scan labeled DECT

WHAT IT ENCODES

Dual-energy CT exploits the differing X-ray attenuation of urate to color-code and quantify MSU deposits noninvasively, making it useful when aspiration is not feasible or for mapping disease burden. It is sensitive in established gout but can miss very early disease and may produce artifacts (e.g., nail beds, skin).

BOARD TRAP

WRONG answer: treating DECT as the definitive first-line diagnostic that replaces aspiration. Joint aspiration with crystal analysis remains the gold standard and the only way to exclude infection — DECT is an adjunct, especially when aspiration is impractical; the discriminator is that imaging cannot rule out coexisting sepsis.

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12. X-ray punched-out erosion

WHAT YOU SEE

Joint X-ray labeled X-RAY

WHAT IT ENCODES

Chronic gout shows 'punched-out' (rat-bite) erosions with sclerotic, OVERHANGING margins and characteristically PRESERVED joint space and bone density until late — distinct from the uniform joint-space loss and periarticular osteopenia of rheumatoid arthritis. Soft-tissue tophi may be visible. Early acute gout typically shows only soft-tissue swelling.

BOARD TRAP

WRONG answer: reading cartilage calcification (chondrocalcinosis) as a sign of gout. Chondrocalcinosis points to CPPD, whereas gout shows punched-out erosions with overhanging margins and preserved joint space — the discriminator is erosive bone change with overhanging edges vs calcified cartilage.

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13. Chalky white tophus

WHAT YOU SEE

Label 'CHALKY WHITE PASTY' near the desk

WHAT IT ENCODES

Aspirated or extruded tophus material is chalky-white and pasty (toothpaste-like), composed of densely packed MSU crystals in a protein matrix. Microscopy of this material shows abundant negatively birefringent needles, confirming chronic tophaceous gout.

BOARD TRAP

WRONG answer: confusing chalky-white tophaceous material with the purulent fluid of septic arthritis or the calcific paste of apatite disease. The discriminator is polarized microscopy: tophus paste is loaded with negatively birefringent urate needles, not bacteria or alizarin-red-staining apatite.

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14. IL-6 pickpocket

WHAT YOU SEE

Magnifier over 'IL-6 PICKPOCKET' figure

WHAT IT ENCODES

Acute-phase reactants (CRP and ESR, driven by IL-6) rise during a gout flare but are NON-specific — they reflect inflammation of any cause and cannot diagnose gout or distinguish it from infection. Note the inflammatory cascade in gout is largely IL-1beta/NLRP3 inflammasome driven, which is why IL-1 antagonists (e.g., anakinra, canakinumab) work in refractory flares.

BOARD TRAP

WRONG answer: using an elevated CRP/ESR (or even a high WBC) to confirm gout or to rule out septic arthritis. These markers are non-specific and rise in infection too — the discriminator remains synovial crystal analysis plus Gram stain/culture, not acute-phase reactants.
